Square of Opposition and Logical Layouts

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1 Introduction

The square of opposition is a diagram illustrating the four syllogistic forms and their logical relationships espoused by Aristotle (384-322 BCE). Logical layouts are diagrams of logic defined by a grammar of the draw operator. The motivation to pair the square of opposition with logical layout was to study and illustrate the syllogistic forms (a, e, i, o) in an alternate logical format.

2 Construction

Forms of syllogisms are known as: Barbara, Celarent, Darii, Ferio, Baroco, Bocardo, Barbari, Cezaront, Camestros, Felapton and Darapti. Each name indicates the subject-predicate terms and their relationship. Barbara indicates MaP encoding that $all\ M\ is\ P$. The objective is to represent the square of opposition in a logical layout.

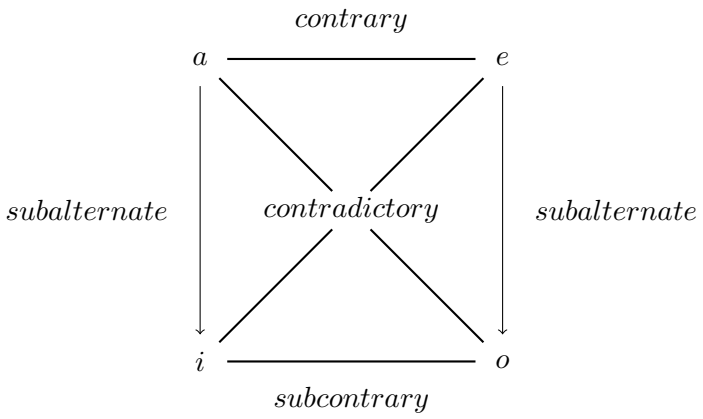


Figure 1: Square of Opposition

Building from the foundation of logical layouts the following forms are then defined in terms of draw operations:

AaB, $All\ A\ is\ B$ is BAP_a where $P_a = T(ea)$

AeB, $No\ A\ is\ B$ is BAP_e where $P_e = F(ea)$

AiB, $Some\ A\ is\ B$ is BAP_i where $P_i = T(oi)$

AoB, $Some\ A\ is\ not\ B$ is BAP_o where $P_o = F(oi)$

The four syllogistic forms can then be represented in one of many ways such as $VHOOHOO, oiea$. Squares of opposition represents the qualities of the forms with relations between them: $contrary$, $subcontrary$, $contradictory$ and $subalternates$.

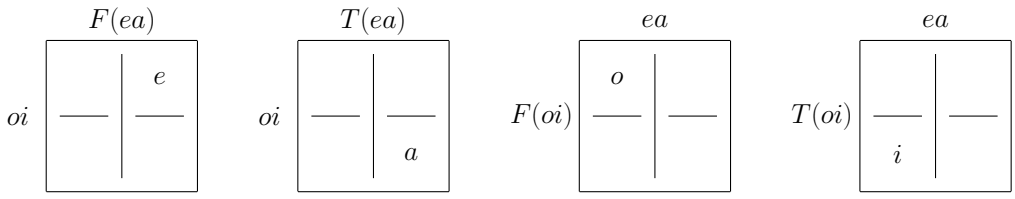


Figure 2: Singular Form Logical Layouts

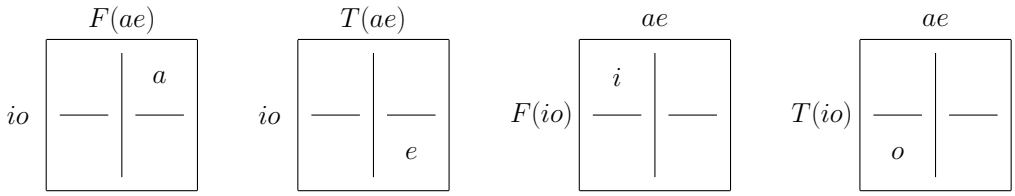


Figure 3: Singular Form Layouts

The logical layout draws on P_0, P_1, P_2 and P_2's $denial$. The simultaneity of the logical layout means the universal $\mathcal{U}$ and particulars $\mathcal{P}$ needs to be considered pairwise. $Not\ P_2$ is o since the predicate is negated. Then P_0 and P_1 are universals. Simultaneously, subalternates P_0 and P_2 are related as are P_1 and $Not\ P_2$. $Not\ P_2$ is contradictory to P_0 so P_0 is a. P_1 is contrary to P_0 so P_1 is e. In the same reasoning, P_2 is subcontrary to $Not\ P_2$ so P_2 is i.

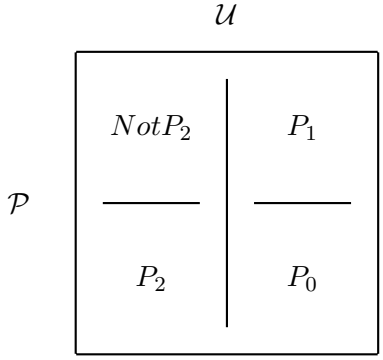


Figure 4: One Possible Logical Layout of the Square of Opposition

Given the first syllogism $Barbara$,

All a is $T(ea)$
All $F(ae)$ is a
$\vdash$
All $F(ae)$ is $T(ea)$

This means the $denial$ of ae is the *affirmative* of ea. In the discourse of logical layouts, the square of opposition relates both ae and ea as $universals$. Two different logical layouts are defined that are then shown to be identical using the $square\ of\ opposition$.

All e is $F(ea)$
All $T(ae)$ is e
$\vdash$
All $T(ae)$ is $F(ea)$

This second syllogism is similar to the first. The $denial$ of ea is *affirmation* of ae. Note that the difference between the $major$ premises of the first two syllogism is the $denial$ or *affirmation* of ea. The $denial$ or *affirmation* of ea is $contrary$.

All i is $T(oi)$
All $F(io)$ is i
$\vdash$
All $F(io)$ is $T(oi)$

The third syllogism is related to the $particulars$ io. Descriptions of the first two syllogisms are descriptive of the last two. The $denial$ of io is the *affirmative* of oi.

All o is $F(oi)$
All $T(io)$ is o
$\vdash$
All $T(io)$ is $F(oi)$

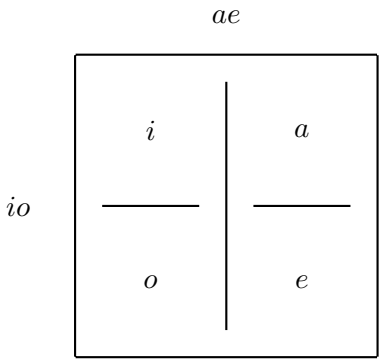


Figure 5: Logical Layout of the Forms $ioae$

The first and third syllogisms relate ea to oi, and the second and fourth relates ae to io. As between the first two syllogisms, the $denial$ or *affirmation* of io is but $subcontrary$. With each form and $Barbara$, it can be concluded that the layouts of figures 5 and 6 are equivalent.

$Subalternates$ are from $F(ae)$ to $F(io)$, and also from $T(ae)$ to $T(io)$.

$Contradictions$ are the simultaneous relations $T(ae)$ and $F(io)$, as well as, $F(ae)$ and $T(io)$.

The logical layout draws on P_0, P_1, P_2 and $Not\ P_2$. The one possible layout in figure 4 associates $Not\ P_2$ with the form o in figure 6.

$Unary$ operators are defined on logical layouts using the $draw$ operator.

$Fallacy$ is defined by $\mathcal{F}_\square = \{P_\square, P_F = F, P_T = F\}$.

$Identity$ is $\mathcal{I}_\square = \{P_\square, P_F = F, P_T = T\}$.

$Negation$ is $\mathcal{N}_\square = \{P_\square, P_F = T, P_T = F\}$.

$Tautology$ is $\mathcal{T}_\square = \{P_\square, P_F = T, P_T = T\}$.

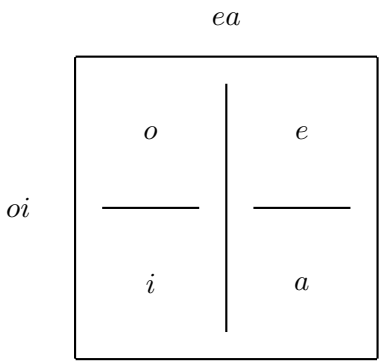


Figure 6: Square of Opposition in the form of a Logical Layout

A is $Tautology$.
E is $Negation$.
I is $Identity$.
O is $Fallacy$.

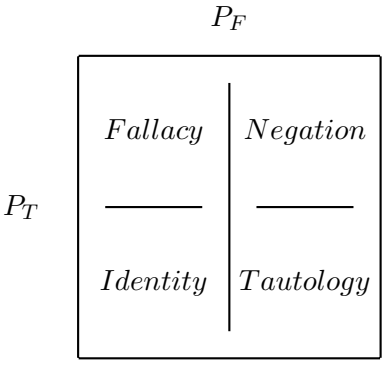


Figure 7: Logical Layout of Unary Operators

3 Conclusion

The square of opposition is an illustration of the relationships of the four forms, $aeio$. Logical layout definition begins with a general description using predicates P_0, P_1, P_2 and $Not\ P_2$. With the help of the $Barbara$ syllogism, the logical layout of the square of opposition is realized. Finally, the four forms $aeio$ are related to the $unary$ operators by shared positions in the logical layout. Further research is warranted.